Hot Days Turn Deadlier Where Guns Are Easy to Carry

Lessons from "Access to Guns in the Heat of the Moment: More Restrictive Gun Laws Mitigate the Effect of Temperature on Violence" by Jonathan Colmer & Jennifer L. Doleac in *The Review of Economics and Statistics*, 2026.

Hot days lead to more gun homicides, but stricter concealed-carry laws cut these increases from 6% to 2%. On an average day, one degree of extra heat means more homicides nationwide — except in states that restrict who can carry a concealed gun, where that heat-driven increase lessens.

BACKGROUND

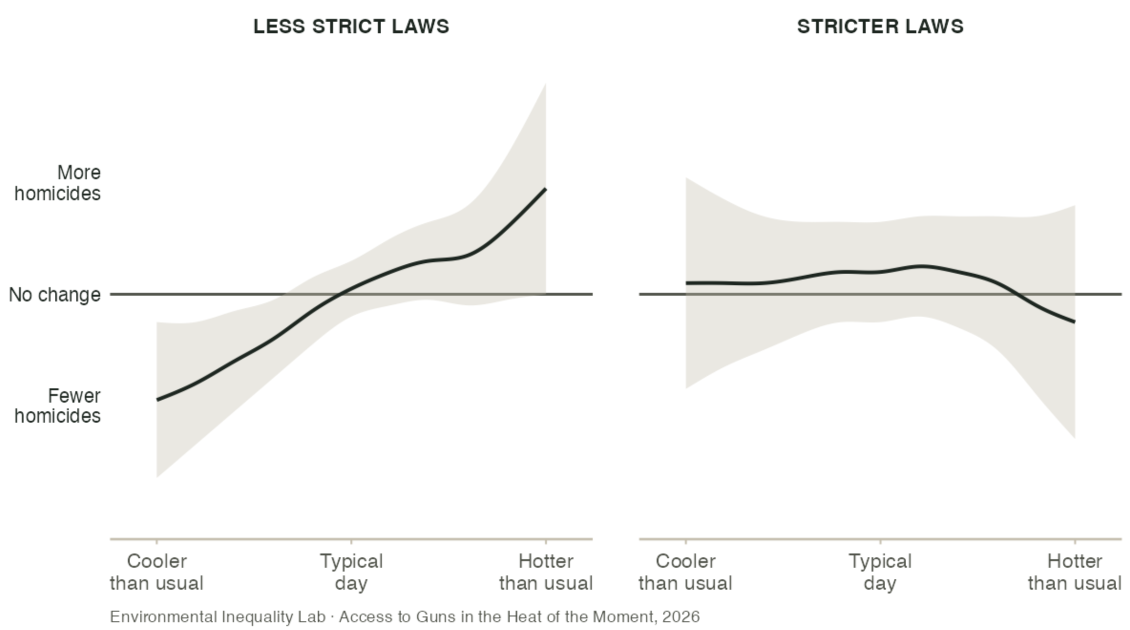
Past research has established that high temperatures lead to increased aggression and violent behavior: hotter days are linked to more confrontations and violent crime. Additionally, states differ significantly in their **concealed-carry laws** — the rules governing who can carry a hidden handgun in public. Some states let nearly any adult carry one; others require a permit that officials can deny, or ban concealed carry altogether. Together, these two facts raise an important question: if heat makes people more likely to become violent, does having a gun close at hand change the results of these confrontations? If so, then stricter concealed-carry laws may protect against an increase in homicides as temperatures rise.

To answer this question, researchers matched 25 years of daily police crime reports (1991–2016) with local weather records and each state's concealed-carry laws, month by month. This allowed them to see the temperature, time, and laws in place for each crime that was committed.

WHAT THE STUDY FOUND

HOW MUCH HOTTER DAYS RAISE HOMICIDES, DEPENDING ON HOW STRICT A STATE'S GUN LAWS ARE

Source: Colmer & Doleac (2026).



Note: Each panel shows the estimated relationship between daily temperature and homicides, after accounting for typical seasonal and local patterns. In states with less strict concealed-carry laws (left), hotter days bring more homicides. In states with stricter laws (right), the line stays flat — hotter days aren't any more dangerous. The shaded areas show the range of uncertainty around each estimate.

It is difficult to measure how a law might affect heat-induced gun violence because it is impossible to measure the difference between an area if the law had never been passed in the first place. The reason the law was passed might also affect gun habits or violent behavior. It would also be difficult to compare states with different laws and/or temperatures for the same reason: there might be other circumstances (e.g. cultural, political, poverty) that would lead to different outcomes that could also affect violence.

However, researchers found a solution that allowed them to judge the actual effects they wanted to observe. They compared the amount of *extra* violence a hot day brought in strict-law places versus loose-law places compared to each state's own averages — shown in the figure above. The researchers assumed that whatever else makes strict-law and loose-law states different — their politics, policing, or poverty rates — doesn't also happen to change how people respond specifically to hot weather. Under that assumption, their comparison isolates how much a state's gun laws shape hot weather's effect on violence, rather than picking up other differences between the states. As a check, they also tracked the same states over time, comparing each state's own hot-day homicides before and after it changed its own law. Both approaches point to the same conclusion.

Nationally, each one-degree-Celsius increase in daily temperature is tied to roughly six percent more homicides than usual on an average day. In states with stricter concealed-carry laws, that heat-driven bump nearly disappears — both when comparing different states at the same time, and when tracking the same states before and after they changed their own laws. **Stricter concealed-carry laws blunt most of the effect of heat on homicides.**

This pattern is also concentrated specifically in gun homicides — not knives or other weapons, and not aggravated assaults broadly. That suggests strict laws aren't stopping hot-day arguments from happening in the first place; they're making the arguments that do happen less likely to turn deadly.

TAKEAWAYS

A GROWING RISK

Concealed-carry laws have loosened nationwide since the 1990s — from 36 states with strict laws in 1991 to just 10 by 2016. Because of rising temperatures from climate change, this effect is only growing. If the whole country still had 1991's stricter mix of laws today, a one-degree hot day would cause an estimated 97 additional homicides nationwide; with today's looser laws in place, that same hot day causes roughly 299. **Fewer states now have the laws that lessen heat's effect on violence, just as rising temperatures make that protection more important.**

PUTTING A PRICE ON A HOT DAY

Using standard estimates of the cost of a homicide to society (about $11.3 million each), the authors calculate that avoiding these heat-driven deaths nationwide would be worth roughly $3.4 billion for every one-degree-Celsius rise in temperature — comparable to the estimated social benefit of hiring nearly 13,000 more police officers. The authors are clear this is a rough, illustrative calculation, not a precise cost-benefit accounting. **Even by conservative estimates, the stakes of heat-driven gun violence are large.**

AGGRESSION, NOT MORE ENCOUNTERS

Why does heat make hot days more lethal where guns are easy to carry? It could be that people simply see more people when it's sunny outside, or that heat makes people more aggressive once a conflict starts. The evidence favors the latter. The effect holds for homicides at home when they aren't necessarily seeing more people, and the effect is strongest in the hottest hours of the day, which doesn't point to an increase in social interactions as strongly. **The heat makes people more aggressive, which turns an ordinary conflict into a lethal one when guns are nearby.**

WHAT THIS DOES (AND DOESN'T) SHOW

This study is about impulsive, heat-driven homicides. It doesn't speak to premeditated crime or gun violence more broadly. It also doesn't weigh the costs to gun owners of more restrictive laws, or say whether stricter concealed-carry laws should be adopted; that tradeoff is a policy judgment the authors leave open. **What the research does show is that temperature is a real, measurable driver of gun violence — and how much it matters depends on the laws already in place.**